

Econometric Methods - Final project

Research Topic #4



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Introduced to empirical economics by the seminal works of Ashenfelter (1978) and Ashenfelter and Card (1985), Differences-in-Differences (DiD) estimator was described as “probably the most widely applicable design-based estimator” by Angrist and Pischke (2010). This indeed widely used estimator aims at estimating the Average Treatment effect on the Treated (ATT) by taking the difference in the difference in outcomes between a treated group and control group, before and after treatment delivery. As emphasized by Kahn-Lang and Lang (2020), this estimator is in fact a solution for a lack of a randomized experimental design. As a matter of fact, when the treatment assignment is truly random, the treated and control group are identical in baseline characteristics and a simple difference in outcomes between the two groups after the treatment delivery identifies the ATT. The problem is that in practice, such experiment rarely naturally occurs, hence the success of DiD in empirical economics. While this estimator allows difference in levels between the two groups by design, ATT identification however requires the control group to be a good counterfactual in trend for the treated group after the treatment delivery. In other words, trends in outcomes followed by the two groups must be “parallel”. If the assumption that the control group is an appropriate counterfactual in the treatment period is not testable, researchers can test that trends across the two groups were at least parallel in the period preceding the treatment. The null hypothesis of such test is the absence of “pre-trends”, or in other words, that trends were parallel. A rejection of the null calls for caution in interpreting the estimated ATT, which is potentially biased. Pre-trend tests became common practice (Bilinski and Hatfield, 2018; Roth, 2022) along with the thriving of DiD, particularly in the 2010s’, and failing to reject the null a requirement to be published in most journals (Roth, 2022).

Nevertheless, a recent literature has emphasized the limits of pre-trend tests. There is a first body of literature contesting such test as being a correct approach for proving that the control group is a good counterfactual in the treatment period. Kahn-Lang and Lang (2020) emphasize that

a difference in levels is an experimental failure and that researchers must provide an economic intuition for why there would be a difference in levels and not in trends. They add that ruling out pre-trends is neither necessary nor sufficient for the parallel trends condition to hold. Moreover, they recall that failing to reject the null does not even rule out pre-trend, it simply fails to prove that there were some. This point is especially developed in Bilinski and Hatfield (2018), who recall that failing to reject the null may be a type II error (false negative), which has to do with the power (" $1 - \beta$ ") of the test and not with the " α " — the probability of rejecting the null while trends were in fact parallel. The power of such test is almost never reported, and found to be in average very low (Ioannidis et al., 2017). Consistent with this literature contesting the pre-trend test approach, another recent body of literature focus on developing robust methods to parallel trend violations. These include instrumenting a confound with a covariate uncorrelated with the treatment (Freyaldenhoven et al., 2019) or interpolating pre-trend of various functional forms in the treatment period and evaluate the sensitivity of the estimated ATT to these various specifications (Rambachan and Roth, 2023).

The aim of this project is to get a better understanding of the criticisms addressed to parallel trend tests by documenting the finite-sample rejection probabilities of the standard t-test of the canonical (homogenous treatment effect and timing) DiD estimand of the ATT over 1000 Monte-Carlo simulations. Our primary focus will lie on over-rejection (type I error) probabilities of this test, i.e. when the t-test rejects the null of no treatment effect when the underlying DGP implied no treatment effect but potentially violated parallel trends. We will consider several standard approaches found in the literature for testing the parallel trend assumptions and for estimating the ATT and compare them in view of over-rejection probabilities of the non-significance of the ATT.

We compare various procedures found in the literature, and show that the most robust ones to parallel trend violations – the event-study specification and rejecting the parallel trend assumption whenever a pre-trend coefficient is individually significant – are also the less powerful in detecting a non-zero ATT.

Section 2 presents our methodology. Section 3 documents over-rejection probabilities of the standard t-test of no treatment effect. Section 4 focuses on the power of this test. Section 5 concludes.

2 Methodology

2.1 Data Generating Processes

We consider a panel of $n = 200$ individuals and $T = 10$ periods, in which a treatment is randomly assigned across the n individuals and delivered during the 5th period for all individuals, normalized to $t = 0$, and further referenced as the baseline period. The outcomes y_{it} are generated according to the following data-generating process (DGP):

$$y_{it} = (\alpha + \beta D_i) + (\gamma + \delta D_i) \cdot t + \sigma D_i \cdot \mathbf{1}_{t>0} + \varepsilon_{it} \quad (1)$$

where α is the intercept for the control group, β shifts the intercept for the treated group, so that treated units have intercept $\alpha + \beta$, γ represents the common linear time trend shared by both groups, D_i is a binary indicator equal to 1 if individual i is treated, 0 otherwise. δ captures a divergence in slopes across the two groups: the parallel trend assumption is violated when it is different from 0. σ captures the treatment effect of a treatment delivered in $t = 0$. ε_{it} are idiosyncratic errors drawn independently from a normal distribution: $\varepsilon_{it} \sim \mathcal{N}(0, 1)$.

In this project, we set α, β, γ to 1 and consider various values for δ and σ depending on the specification, allowing us to model a linear monotonic violation of parallel pre-trends, and a homogenous treatment effect in the form of a discrete jump.

Note that (1) is the assumed underlying DGP when the canonical (homogenous treatment timing and effect) DiD estimator is used. The multiple periods will allow us to estimate the ATT within the event-study framework, which is a common approach even when treatment effect and timing are homogenous (see e.g., Bosch and Campos-Vazquez, 2014; Deschênes et al., 2017). In particular, we will not consider a staggered DiD framework in the spirit of Callaway and Sant’Anna (2021) and Sun and Abraham (2021) in which the event-study specification is also used. Using this simple setting will serve the purpose of better understanding the grounds of the criticisms addressed to the parallel trend testing approach, whose success coincided with that of the canonical DiD.

2.2 Parallel trend tests

Consistent with the standards of high-ranked journals (Roth, 2022), we test the parallel trend assumption before estimating the treatment effect. As explained earlier, this test consists in practice in assessing whether this assumption held before the treatment. Let H_0^{pre} denote the null hypothesis that the pre-treatment trends are identical for the treated and control units. The test is considered to be successfully passed if the authors fail to reject H_0^{pre} . Again, this can be because trends were indeed parallel, but it can also be a type II error (Bilinski and Hatfield, 2018;

Kahn-Lang and Lang, 2020; Roth, 2022), a point we will go back to.

Concretely, we estimate the following event-study specification, i.e. for all $i \in \llbracket 1, n \rrbracket$, with $t = 0$ as baseline:

$$y_{it} = \beta \cdot D_i + \lambda_t + \sum_{\tau \neq 0} \delta_\tau \cdot D_i \cdot \mathbf{1}_{t=\tau} + \varepsilon_{it}, \quad (2)$$

where β captures the difference in levels between the two groups, λ_t are time fixed effects, D_i is a binary indicator equal to 1 if individual i is treated, and $\mathbf{1}_{t=\tau}$ is a dummy variable equal to 1 at year τ . Each coefficient δ_τ captures the difference in trends across the two groups at time τ : when $\tau > 0$ it captures the treatment effect. The error terms ε_{it} are assumed to be i.i.d. normal. Consistent with the common practice found in the literature, we test the null hypothesis: $H_0^{\text{pre}} : \delta_{-4} = \dots = \delta_{-1} = 0$.

Two ways of testing the joint insignificance of the p pre-trend coefficients are found in the literature: **Standard procedure:** The most common practice is to refer to the two-sided t-test associated with each pre-trend coefficient found in the regression output, and to reject the null if $\exists \tau < 0$ such that:

$$\left| \frac{\hat{\delta}_\tau}{\widehat{\text{SE}}(\hat{\delta}_\tau)} \right| > t_{1-\alpha/2}^{(N-p)}$$

where $t_{1-\alpha/2}^{(N-p)}$ is the $(1 - \alpha/2)$ quantile of the Student distribution with $N - p$ degrees of freedom and α a given threshold, generally 5%.

Multiple hypothesis test: Less widespread and yet more appropriate tools on paper, multiple hypothesis tests – Fischer, Wald, Score, Likelihood Ratio – can also be used (see e.g. Courtemanche et al., 2017; Yurukoglu et al., 2017; Goldman et al., 2018 for the F-test). In this project, we will compare the standard method to a Fischer test, but since we are considering a linear model with iid normal errors, the other multiple hypothesis tests would yield similar results as the F-test. Such test compares the restricted model (excluding the $\{D_i \cdot \mathbf{1}_{t=\tau}\}_{\tau \leq 0}$ interaction terms) with the unrestricted model and rejects H_0^{pre} when:

$$F = \frac{(SSR_r - SSR_u)/p}{SSR_u/(N - k)} > F_{1-\alpha}^{p, N-k}$$

where SSR_r and SSR_u are the sum of squared residuals from the restricted and unrestricted models respectively, N is the total number of observations (i.e., $N = n \cdot T$), k the number of parameters in the unrestricted model and $F_{1-\alpha}^{p, N-k}$ the $(1 - \alpha)$ quantile of the Fischer distribution with $d_1 = p$ and $d_2 = N - k$ degrees of freedom.

Bonferroni-adjusted standard procedure: To better understand how the standard parallel trend test and the multiple hypothesis test compare, we consider a third alternative test. What is called Bonferroni correction was described in Dunn (1961) and links individual and multiple hypothesis testing by stating that individual hypothesis should be tested at a significance level of $\frac{\alpha}{p}$ to compensate for the increased risk of incorrectly rejecting H_0^{pre} when the number of tests increases. In our case with 4 pre-periods and an overall desired over-rejection probability of 5%, the corrected threshold becomes: $\alpha_{\text{Bonf}} = \frac{0.05}{4} = 0.0125$. We will check that once this threshold is adjusted, the standard method becomes fairly equivalent to the Fischer test.

2.3 Main test: significance of the ATT

Unless we estimate the ATT *unconditionally* on the pre-trend test outcome for comparison purposes, we will accept $H_0 : \text{ATT} = 0$ whenever H_0^{pre} is rejected when using one of the three tests previously described. *Conditional* on passing the parallel trend test, i.e. on failing to reject H_0^{pre} , we propose to estimate the ATT using the canonical DiD estimator in two different ways. The first approach is the textbook DiD specification:

$$y_{it} = \beta \cdot D_i + \lambda_t + \sigma \cdot D_i \cdot \mathbf{1}_{t>0} + \varepsilon_{it}, \quad (3)$$

which implicitly constrains the pre-trend coefficients to be zero by excluding them from the model. This model will be referred to as the *constrained model*. We reject H_0 if the p-value associated with the t-test of the significance of σ is below $\alpha = 5\%$.

Alternatively, we estimate an event-study specification which became widespread in the more recent years as it allows to assess the parallel trend assumption while estimating the ATT:

$$y_{it} = \beta \cdot D_i + \lambda_t + \sum_{\tau \neq 0} \sigma_{\tau} \cdot D_i \cdot \mathbf{1}_{t=\tau} + \varepsilon_{it} \quad (4)$$

(4) does not constrain the pre-trend coefficients to be zero. This is why we will refer to this specification as the *unconstrained model*. The ATT is estimated by averaging the post-treatment coefficients: $\widehat{\text{ATT}} = \bar{\sigma}_{\text{post}} = \frac{1}{5} \sum_{\tau=1}^5 \hat{\sigma}_{\tau}$, and the Delta Method is used to estimate its standard error.

In the following sections, we compare these various procedures for testing parallel trends and estimating the ATT, in particular in view of the finite-sample over-rejection probabilities of no treatment effect.

3 Over-rejection probabilities of no treatment effect

In this section, we consider various DGPs which all imply no treatment effect ($\sigma = 0$) but differ in terms of pre-trends. That is, we consider a grid of pre-trends $\{\delta_m\}_{m=0}^{16}$ with $\delta_m = -0.4 + \Delta \cdot m$ and $\Delta = 0.05$ and run for each δ_m 1000 Monte-Carlo simulations.

3.1 Parallel trend tests comparison

In a first step, we propose to compare the three procedures for testing the parallel trends: the standard way, the Fischer test, and the Bonferroni-adjusted standard test. For each procedure, the rejection probabilities of the pre-trend test and of the main test are respectively plotted in Figures 1 and 2 against $\{\delta_m\}_{m=0}^{16}$.

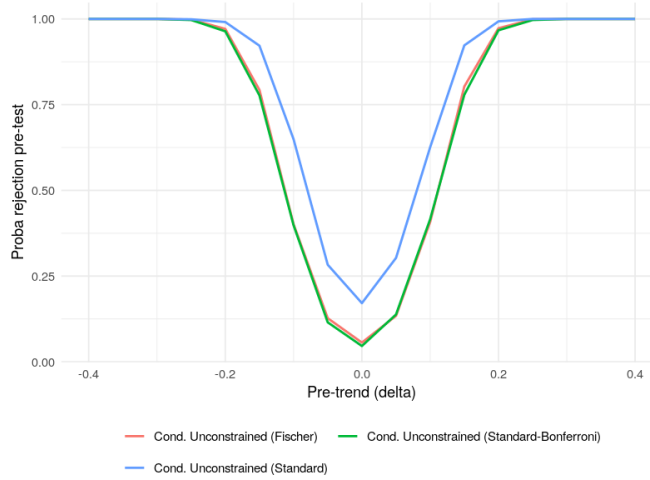


Figure 1 - PARALLEL TREND TEST REJECTION PROBABILITIES

Before considering the over-rejection probabilities of the main t-test, we propose to analyze the rejection probabilities of the parallel trend test. First, we can observe on Figure 1 an over-rejection (type I error) probability of the parallel trend test – the rejection rate when $\delta = 0$ – of 5% when the Fischer test (orange curve) with $\alpha = 5\%$ is used. This is reassuring: the test is well calibrated under the null. Second, we can observe that the standard procedure, which consists in rejecting the null when there exists $\tau < 0$ such that the individual p-value associated to the t-test of significance of $\hat{\delta}_\tau$ is below α , leads to more type I errors than the multiple hypothesis Fischer test. In particular, it over-rejects at a rate of 20%, which is four – the number of pre-trend coefficients – times α , consistent with the Bonferroni correction literature (Dunn, 1961). We check that when "Bonferroni-adjusted", the standard way of testing the parallel trend (green curve) leads to an over-rejection rate of 5%. Therefore, when researchers have multiple pre-treatment periods,

they must adjust the confidence interval size by decreasing it proportionally to the number of pre-treatment periods in order to assess the plausibility of the parallel trend assumption. Otherwise, they significantly increase the risk of rejecting the null when pre-trends were in fact parallel.

However, when the parallel trend assumption is violated ($\delta \neq 0$), the standard pre-trend test (blue curve) exhibits more power than the other procedures. Consider the case where $\delta = 0.15$: the standard procedure rejects H_0^{pre} at a rate of 92% vs. 80% for the other procedures. Therefore, it appears to be a trade-off between power and robustness. On the one hand, the standard approach multiplies the probability of failing the robustness check when it should have passed it (type I error) by the number of pre-treatment periods. On the other hand, the multiple hypothesis test increases the probability of failing to reject H_0^{pre} when it should have (type II error). Bilinski and Hatfield (2018) argue that when it comes to testing assumption such as parallel trends, type II errors are more problematic than type I errors. This would advocate in favor of the non-adjusted standard procedure.

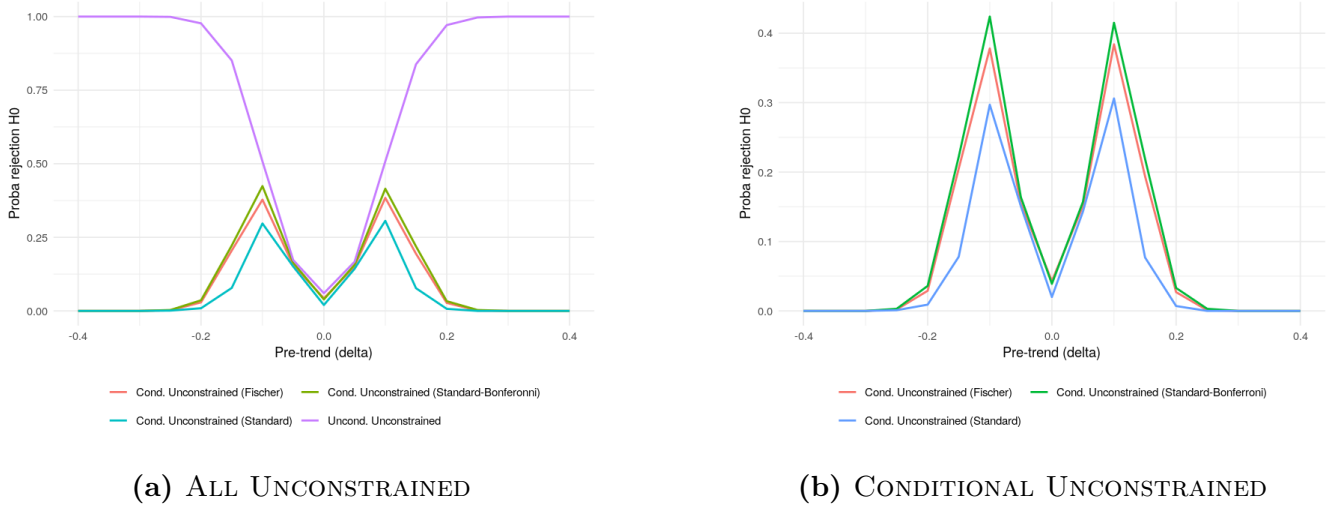


Figure 2 - OVER-REJECTION PROBABILITIES OF THE ATT INSIGNIFICANCE: VARYING PRE-TRENDS

Figure 2 demonstrates the importance of testing the parallel trend assumption for the researchers not to conclude that the treatment has an effect when it does not. Figure 2a compares the rejection probabilities of the ATT insignificance conditional and unconditional on passing the parallel trend test. Recall that in all considered scenarios the treatment has no effect, so these probabilities can be interpreted as type I errors. When it comes to concluding whether a treatment has an effect, this type of error is the most problematic. Recall also that if H_0^{pre} is rejected, we

accept $H_0 : \text{ATT} = 0$ in order to limit over-rejection. This explains why conditioning on the success of the parallel trend test limits type I errors compared to the unconditional model (magenta curve) in which the pre-trend confounds with the treatment, and why it became a publication requirement for most journals (Roth, 2022). Note that this does not contradict the observation of Roth (2022) that conditioning acceptance of articles on passing a parallel trend test biases article selection towards papers with fundamentally violated parallel trends but abnormally low differential pre-trends in effect, leading to a prevalence of biased ATT among published papers due to the mean reversion effect. This could be tested by comparing over-rejection probabilities of simulations that pass the parallel trend test and those that do not.

Figure 2b now focuses on the various parallel trend test procedures in the light of the over-rejection probabilities of the main test, conditional on passing each test. Note that when H_0 is over-rejected, the parallel trend test must have failed to reject H_0^{pre} in the first step. When the main test is conditioned on passing the standard pre-trend test (blue curve), the over-rejection probabilities are smaller for any pre-trend value: this is because as seen in Figure 1, this procedure limits type II errors of the parallel trend test. In other words, because this test was more powerful in detecting parallel trend violations, it limits type I errors in the second step, but at the cost of a higher probability of over-rejecting the parallel trend hypothesis.

3.2 Main specifications comparison

We now propose to compare the "constrained" and "unconstrained" models, which respectively refer to the textbook and the event-study specifications of the canonical DiD. The over-rejection probabilities of the main test are plotted for both specifications and using the standard parallel trend test against $\{\delta_m\}_{m=0}^{16}$ in Figure 3. We can observe that restricting the pre-trend coefficients to be zero (constrained model) increases over-rejection probabilities for any value of the pre-trend. This is because imposing zero pre-trend coefficients amplifies the mean reversion effect. Indeed, when parallel trends are violated ($\delta \neq 0$), the simulations that pass the pre-trend test are those in which differential pre-trends happen to be exceptionally small due to random noise. However, there is no reason for this differential trend to remain small in the treatment period, since the DGP includes a systematic differential trend. This leads to a reversion toward the mean. As a result, the estimated ATT captures both the treatment effect (when present) and the mean reversion effect (Roth, 2022). Restricting the pre-trend to be zero therefore amplifies the bias from mean reversion. Therefore, the event-study specification appears to be more robust to parallel trend violations than the textbook DiD.

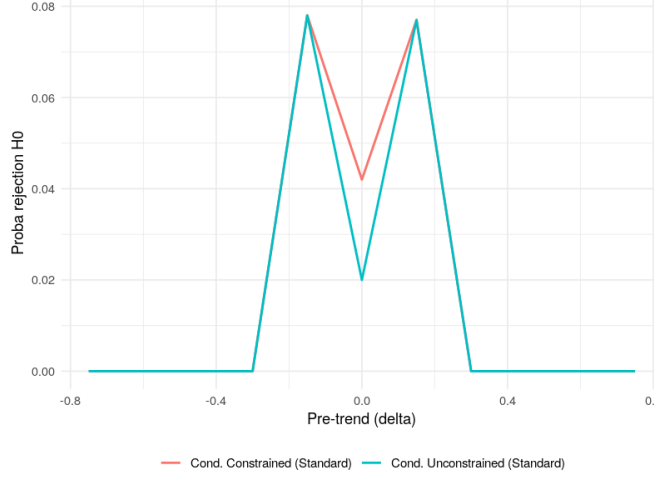


Figure 3 - REJECTION PROBABILITIES OF THE ATT INSIGNIFICANCE: CONSTRAINED VS. UNCONSTRAINED

4 The power to detect a treatment effect

Until now, we have compared various parallel trend tests and DiD specifications in view of their statistical performance in limiting type I errors of the main test. This type of error is the one that journals' editors seek to identify as much as possible in order not to publish papers concluding of a treatment effect when there is not. However, researchers want powerful tests for the ATT significance, capable of detecting an effect when there is one. Using more robust procedures to parallel trend violations may come at a cost of less power. This is what we seek to explore in this section, where we consider parallel trends but various treatment effects.

Figure 4 plots the rejection probabilities of the ATT insignificance against a grid of ATT $\{\sigma_m\}_{m=0}^{10}$

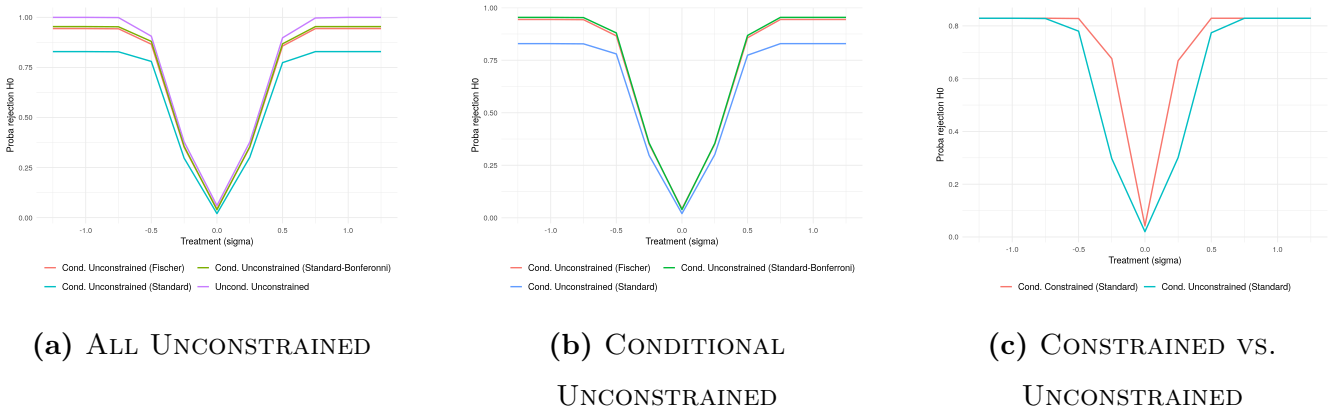


Figure 4 - REJECTION PROBABILITIES OF THE ATT INSIGNIFICANCE: VARYING ATT

with $\sigma_m = -1.25 + \Delta \cdot m$ and $\Delta = 0.25$. Figure 4a compares in particular the main t-test

conditionally and unconditionally on passing the parallel trend test. There is not much of the difference: the gap can only be explained by over-rejection of the parallel trend hypothesis in the first step since trends are now parallel. This explains why using a standard pre-trend test, which produced more type I errors in the first step test, leads to a smaller rejection rate (i.e. more type II errors) of the main t-test. Figure 4c compares the textbook and the event-study DiD specifications. Restricting the pre-trend coefficients to be zero (textbook DiD) gives more power to the main t-test, but recall that this procedure was also less robust to parallel trend violations.

5 Conclusion

We have explored the statistical properties of the main t-test of insignificance of the canonical DiD estimand of the ATT and of the parallel trend test, commonly run as a robustness check. We have compared various procedures found in the literature, and shown that the most robust ones to parallel trend violations – the event-study specification and rejecting the parallel trend assumption whenever a pre-trend coefficient is individually significant – are also the less powerful in detecting a non-zero ATT.

In this project, it should be emphasized that less robust procedures were only weakly robust to small pre-trends (i.e. smaller than one third of the error term standard deviation). One could therefore be tempted to advocate in favor of the more powerful procedures. However, we only have considered linear differential pre-trends, and tested for their presence using a linear parametric form. With real-world data, pre-trends may take different functional forms, unknown to researchers. They implicitly assume them by specifying their model (Kahn-Lang and Lang, 2020). One interesting extension of this project could therefore be to consider various functional forms of differential pre-trends and in particular to test for their presence with an incorrect functional form.

In any case, it is not very satisfactory to be forced to choose between a weakly robust and a weakly powerful procedure, where the latter poses the risk for editors of missing publishable articles. Journals' acceptance standards must evolve. They must in particular demand more economic reasoning, be it for explaining intuitively why trends would be parallel (Kahn-Lang and Lang, 2020), for finding a covariate to instrument a confound (Freyaldenhoven et al., 2019), or for finding a differential pre-trend threshold above which it is considered to be problematic (Bilinski and Hatfield, 2018). They must favor the use of recently developed robust methods to parallel trend violations, including pre-trend interpolations in the treatment period against which the sensitivity of the estimated ATT should be tested (Rambachan and Roth, 2023).

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